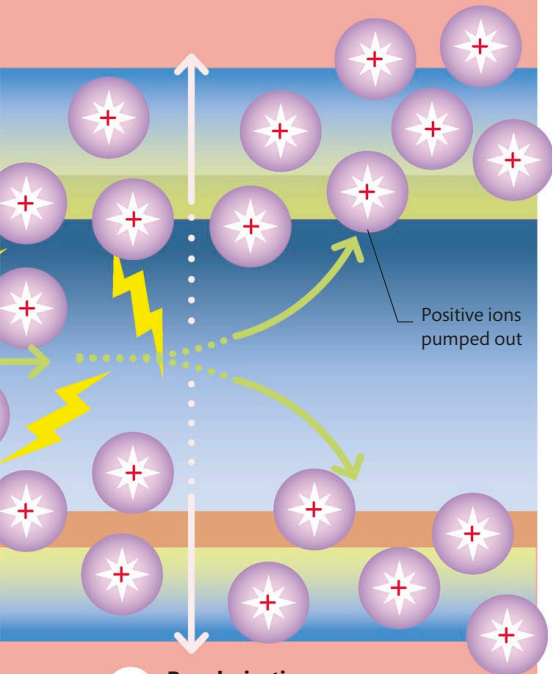




NERVE AGENTS

Chemical weapons, like novichok and sarin, work by interfering with how neurotransmitters behave at the synapse. Nerve agents can be inhaled or act on contact with skin. They prevent the synapse from clearing away used acetylcholine, which is involved in the control of muscles. As a result, muscles, including those used by the heart and lungs, are paralyzed.



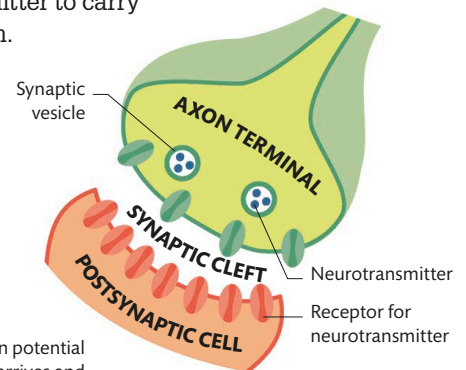
- 3 Repolarization**
The depolarization of a section of the axon causes the neighboring section to undergo the same process. Meanwhile, the cell pumps out positive ions to repolarize the membrane back to the resting potential.

Synapses

Some neurons do not share a physical connection. Instead they meet at a cellular structure, called a synapse, where there is a gap of 40 billionths of a meter, known as the synaptic cleft, between the axon of one neuron (the presynaptic cell) and the dendrite of another (the postsynaptic cell). Any coded signal carried by electrical pulses is converted into a chemical message at the tip, or terminal, of the axon. The messages take the form of one of several molecules called neurotransmitters (see p.24), which pass across the synaptic cleft to be received by the dendrite. Other neurons have electrical synapses rather than chemical synapses. These are effectively physically connected and do not need a neurotransmitter to carry electrical charge between them.

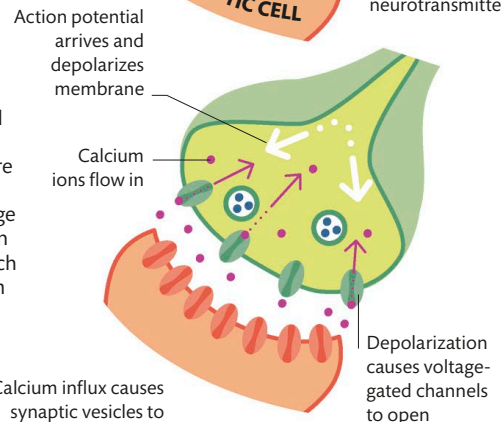
1 Chemical store

Neurotransmitters are manufactured in the cell body of the neuron. They travel along the axon to the terminal, where they are parceled up into membranous sacs, or vesicles. At this stage, the terminal's membrane carries the same electrical potential as the rest of the axon.



2 Signal received

When an action potential surges down the axon, its final destination is the terminal, where it temporarily depolarizes the membrane. This electrical change has the effect of opening protein channels in the membrane, which allow positively charged calcium ions to flood into the cell.



3 Releasing messages

The presence of calcium within the cell sets off a complex process that moves the vesicles to the cell membrane. Once there, the vesicles release neurotransmitters into the cleft. Some diffuse across the gap to be picked up by receptors on the dendrite. The neurotransmitters may stimulate an action potential to form in that dendrite, or they may inhibit one from forming.

